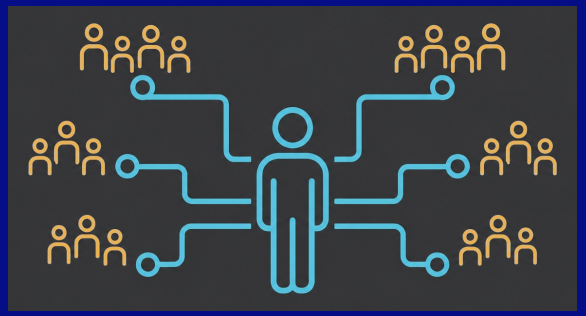


DHUM101
Socially Responsible Computing

University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics

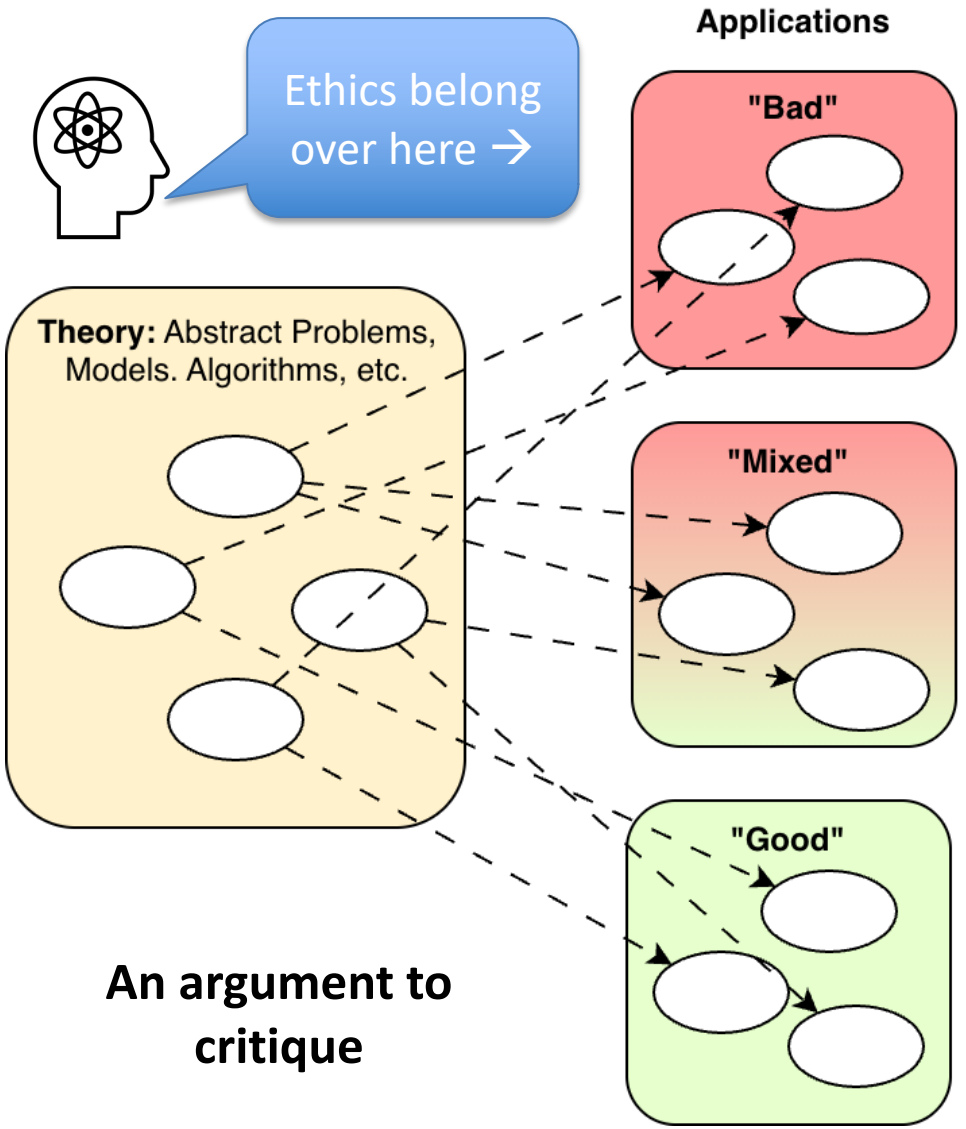
Lecture 10
Is technology neutral? / Computing for social good

- Stephen Cranefield
  Semester 2, 2026

Notices

- Correction from the original slides: We are cancelling Tutorial 11 (not next week's one). Next week's tutorial will be the final one.
- Lab 11 is *not* cancelled.
- Next week's Monday class is an interview with the university's Chief Digital Officer, Steven Turnbull. We want to find out what policies and practices are in place to ensure the university follows responsible computing practices.
  - Please come along. It is rare to have access to high level leaders and policy makers at the university.
  - He has been told that one question we will ask is:
    "Do you have any suggestions for students who would like to have a career pathway that leads to a role such as yours?"
  - **Questions related to this interview could appear in the final test.**

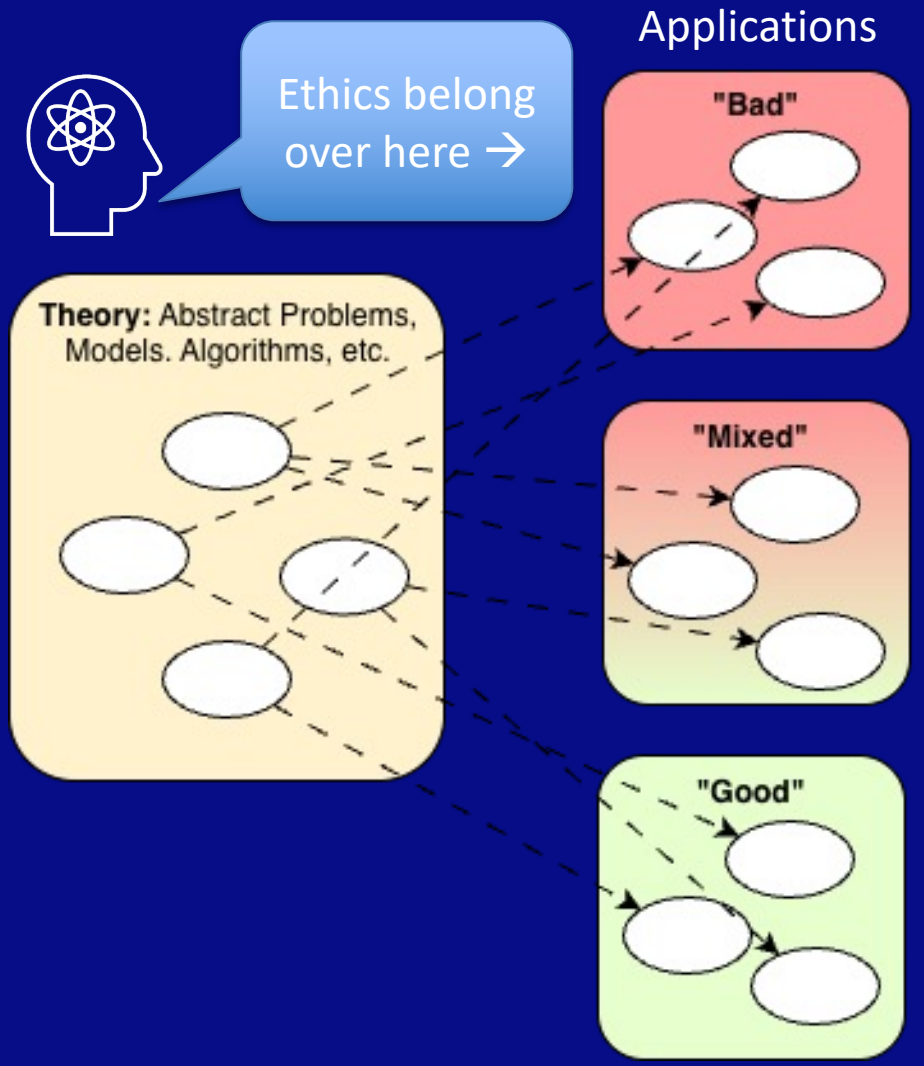
Lecture Overview



An argument to critique

- Are the model and claim on the left a valid way of thinking about when technology can be “good” or “bad”?
- Is there a difference between responsible computing and ”computing for good”?
- How do we define “good”? Is it the intent or the consequences?
- A definition of *computing for social good.*
- Some guidelines for *AI for social good*

Is technology ethically neutral?



An argument based on the *neutrality thesis*

Premises

- Human society grows and prospers through the development of new knowledge.
- The acquisition of knowledge is intrinsically valuable.
- ***Neutrality thesis***: A technology is morally neutral. Its ethical character depends on how it is used.

Apparent conclusions

- Ethics are only relevant when developing *applications* of technology.
- Researchers therefore have no ethical constraints concerning *what* technologies they invent.

Is this argument valid?

Is there something missing?

A missing premise

The responsibility principle

Researchers are responsible for their own actions, not for what others do with the technologies they develop.

Is this true?

Case study 1

In the 1940s, Nazi Germany did not know that a nuclear weapon was possible. Their model and governing equation was wrong, so German theoretical physicist Werner Heisenberg predicted that you would need many tons of uranium to produce a self-sustaining nuclear reaction.

The US had the correct model, which would fit on six pages of A4 paper and led to a prediction that only 60kg of uranium would be needed.

- Suppose you were a German physicist at the time (and not a fanatical supporter of the Nazi Party) and created or learned of the correct model. Would you consider sharing that model to be ethically neutral?
- Do you think the scientists working on the secret US Manhattan Project (to develop nuclear weapons) considered their work to be ethically neutral?
- Is there a difference between these two questions?

Source: "What on Earth is Ethical Maths? With Maurice Chiodo", The Good Robot Podcast, https://www.thegoodrobot.co.uk/listennow/episode/7e08a028/what-on-earth-is-ethical-maths-with-maurice-chiodo

Contrasting opinions from Manhattan Project scientists

- Robert Oppenheimer (Nov. 1945):
  "Because we are scientists … It is our faith and our commitment, seldom made explicit, even more seldom challenged that knowledge is a good in itself. Knowledge and such power as must come with it.".
- The Franck Report (June 1945), secretly written by a group of prominent nuclear physicist and delivered to a committee in Washington DC, proposed *not* to use nuclear bombs for an early unannounced attack on Japan. Instead, they should be "revealed to the world by a demonstration in an appropriately selected uninhabited area". It includes these words:

  "in the past, scientists could disclaim direct responsibility for the use to which mankind had put their disinterested discoveries. We cannot take the same attitude now because the success which we have achieved in the development of nuclear power is fraught with infinitely greater dangers than were all the inventions of the past."
- The Szilárd petition (July 1945) by scientist Leo Szilárd and was signed by 70 Manhattan Project scientists. Based on "moral responsibilities", it called for Japan to be given proper warning and advised of the terms of surrender. It did not reach President Truman.

Sources: https://www.americanrhetoric.com/speeches/robertoppenheimeratomicbomb.htm.
https://www.atomicarchive.com/resources/documents/manhattan-project/franck-report.html, https://en.wikipedia.org/wiki/Szil%C3%A1rd_petition,

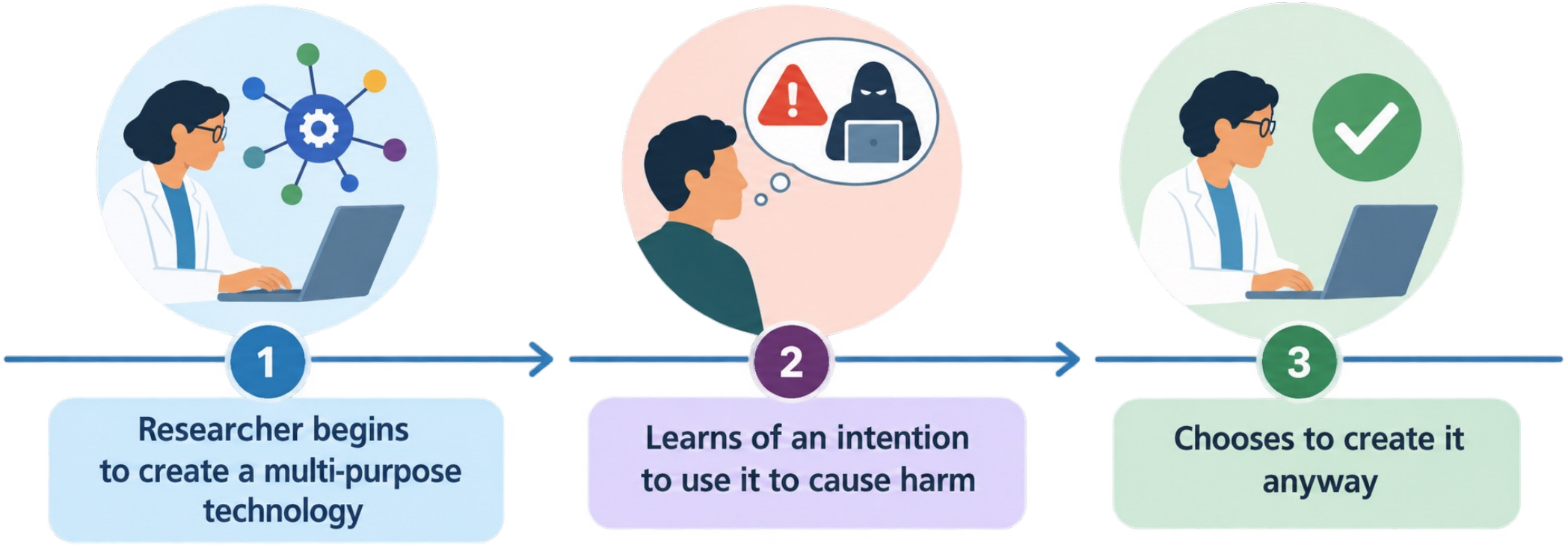
Case study 2: Joseph Redmon

- In 2016, University of Washington PhD student Joseph Redmon, co-developed the YOLO neural network approach for object detection in images. This was "hailed as a milestone in object detection research and led to better, faster and more accurate computer vision [CV] algorithms".
- Excerpts from a 2020 Twitter thread:
  - "I stopped doing CV research because I saw the impact my work was having. I loved the work but the military applications and privacy concerns eventually became impossible to ignore." (Redmon)
  - "I don't quite follow this thought process. If knives are used (by some) to kill people, do we stop producing knives, which also have military applications?" (Someone else)
  - "If you worked in a knife factory and a guy came in and thanked you for making knives because he killed many people with those knives and then he showed you a video of himself killing people with a knife you made how would you feel then about working in your knife factory? (Redmon)

Sources: https://syncedreview.com/2020/02/24/yolo-creator-says-he-stopped-cv-research-due-to-ethical-concerns/amp/, https://x.com/pjreddie/status/1230524770350817280?ref_src=twsrc%5Etfw%7Ctwcamp%5Etweetembed%7Ctwterm%5E1230524770350817280%7Ctwgr%5E%7Ctwcon%5Es1_&ref_url=https%3A%2F%2Fd-2118908205275571450 9.ampproject.net%2F2608131752000%2Fframe.html

What accounts for disagreements about the responsibility principle?

Is choosing to create something, knowing it will likely be used for harm, a *separate* action from creating it before having that knowledge?



Claim 1

Responsibility doesn't stop once you have decided to create a technology. You are taking actions that have moral consequences throughout the development process.

Responsible computing involves the ethical design, development, and use of technology. It emphasizes minimizing harm, respecting user rights, and fostering sustainability. This approach ensures that technology benefits society without creating unintended consequences.

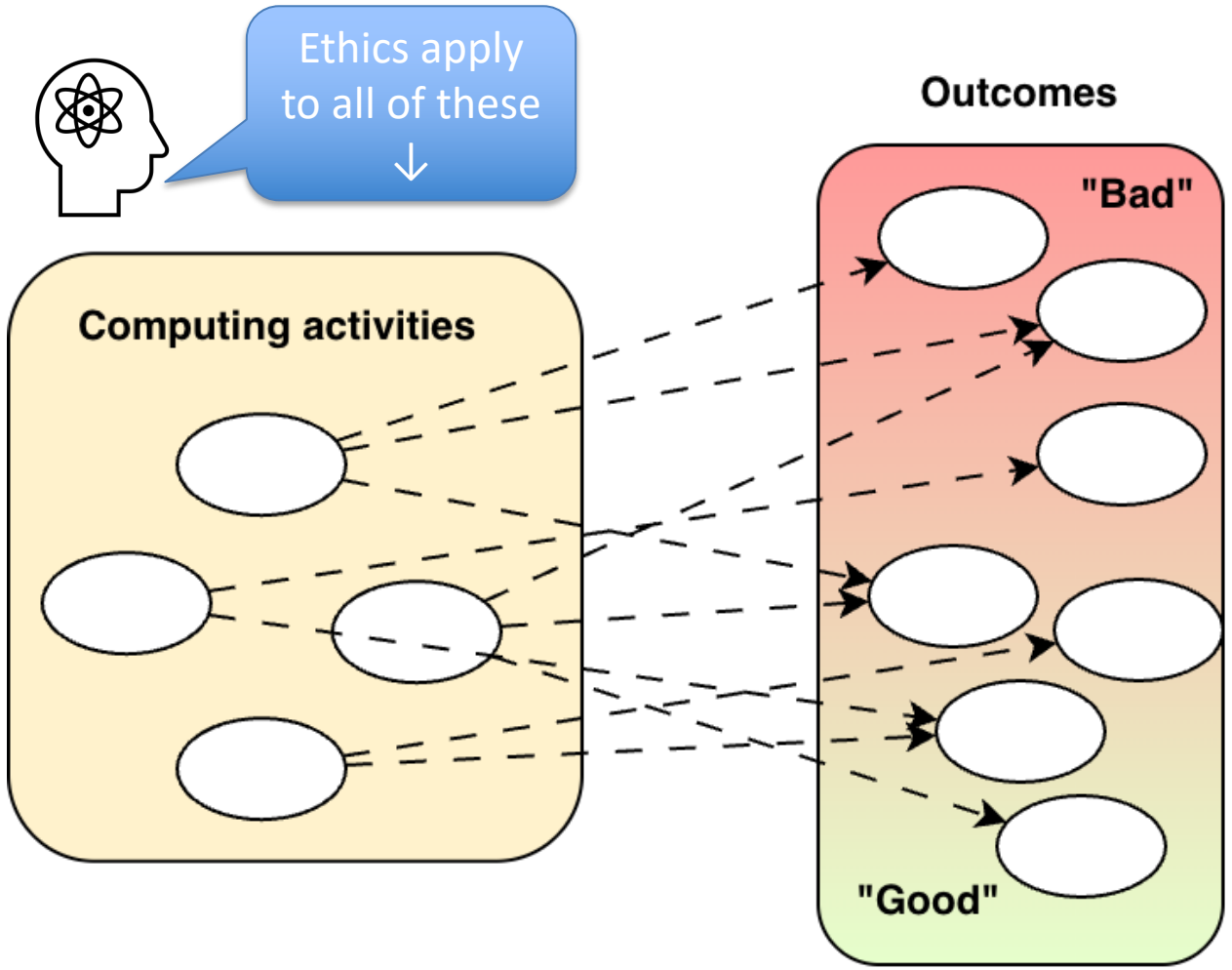
Core Tenets of Responsible Computing

- Ethical Decision-Making: Considering the societal impact of technology at every stage of development.
- Inclusivity: Designing systems accessible to all, regardless of ability, location, or resources.
- Transparency: Ensuring users understand how systems work and how their data is used.
- Sustainability: Reducing the environmental impact of computing through energy-efficient systems.
- Accountability: Holding developers, organizations, and users responsible for ethical breaches.

From Lecture 1

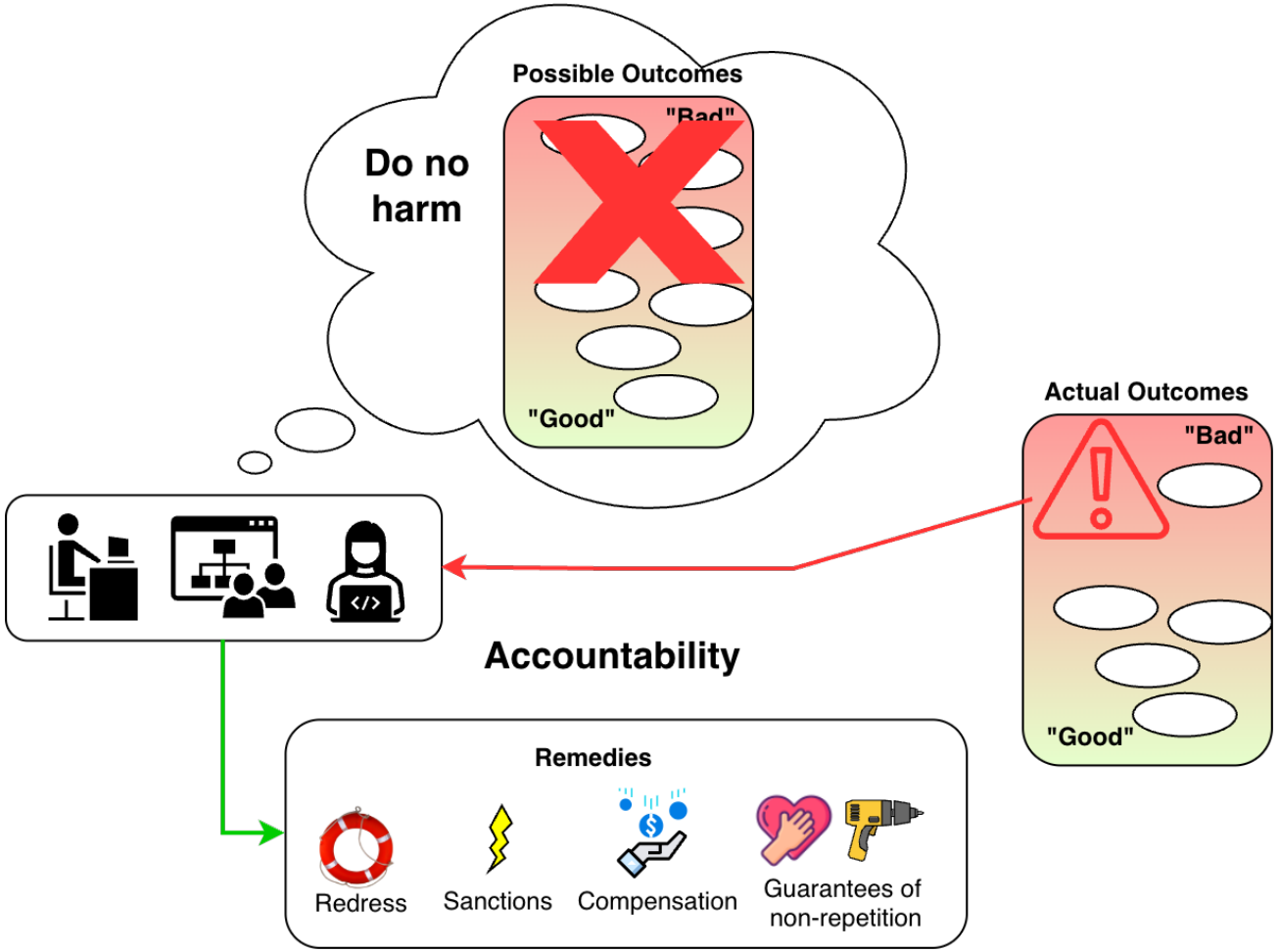
Reproduced with permission from https://www.linkedin.com/pulse/introduction-responsible-computing-dekut-recec-towyf (author Clint Simyu)

A revised model



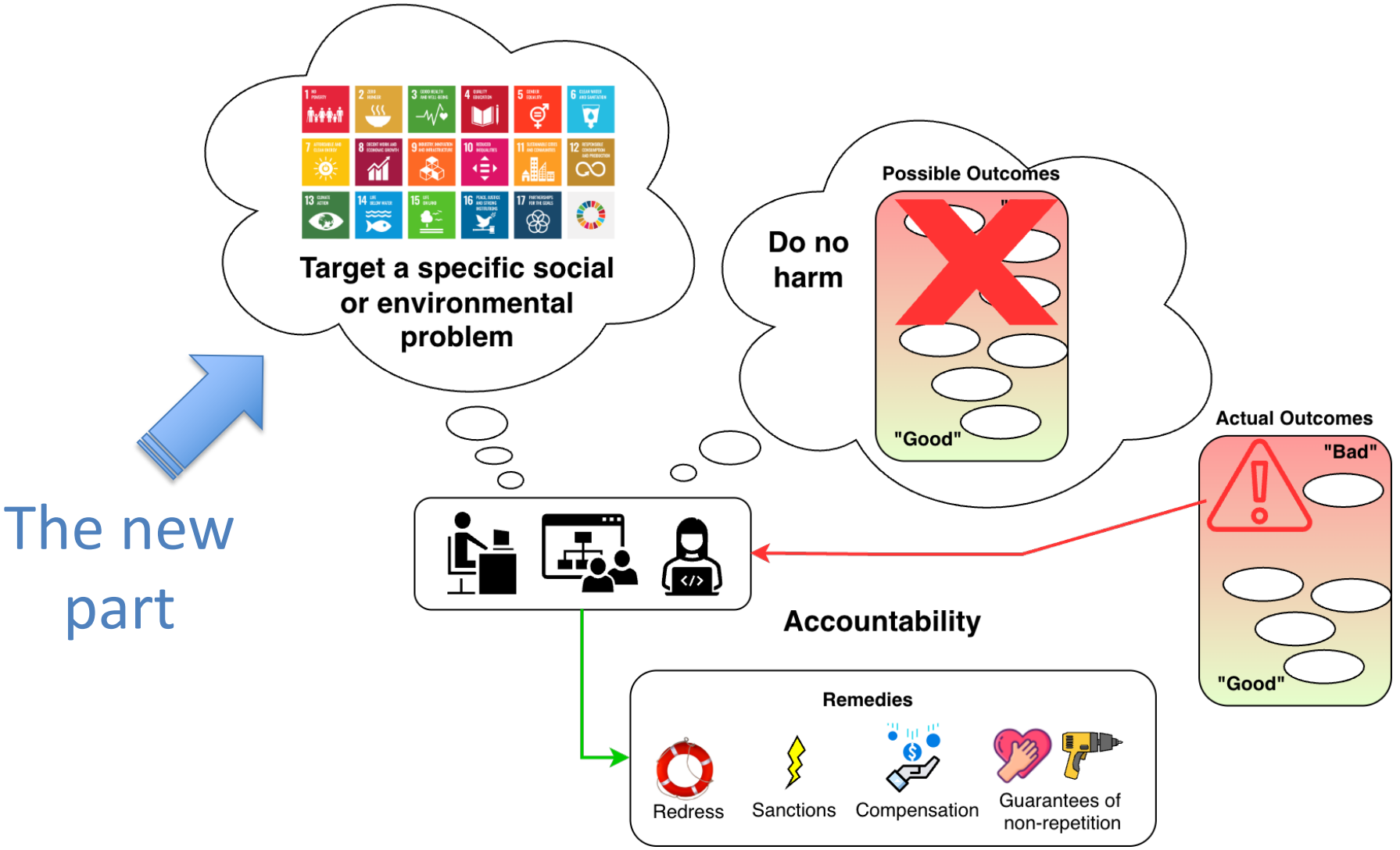
- No distinction made between a technology and its applications
- But does this tell us what responsible computing is?
- Do we measure “good” and “bad” *retrospectively* (consequentialism)?
- Or does the *intent* count?

Intentions matter, but so do bad outcomes



Responsible Computing

In contrast: Computing for Social Good



A definition of Computing for Social Good (CSG)

"CSG is the application of information and computing technology for societal development in areas such as healthcare, education, poverty reduction, and sustainability.

The United Nations Sustainable Development Goals and similar frameworks provide good examples of the scope of CSG.

CSG is computing where the primary intent is to improve people's lives and address societal problems."

CSG does not have to be about creating software. A CSG project could involve collecting and preserving data, building digital infrastructure that is lacking, providing digital services, upskilling community groups to reduce the digital divide, etc.

Source: H.J.C. Ellis et al. "CS2023: Computing for Social Good in Education", ACM Inroads, https://dl.acm.org/doi/epdf/10.1145/3699719

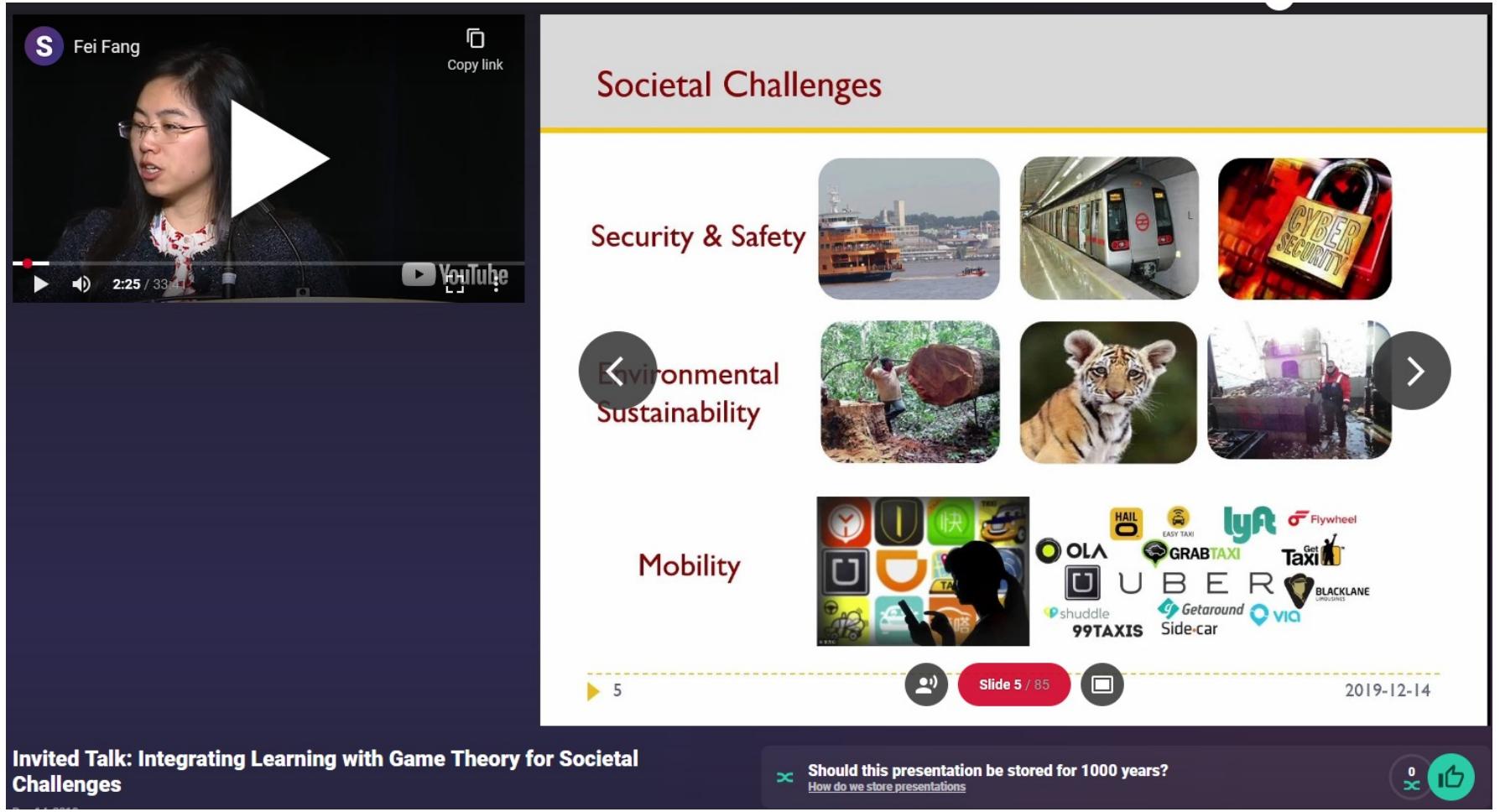
How to go about it?

- Keep those most affected by the project outcomes at the centre of the process.
- The system designer should act as a facilitator, not an expert.
- Prioritise actual community impact over designer intentions.
- Treat everyone as an expert based on their own lived experience and can contribute to the design process.
- Make the process collaborative and accountable.
- Look at what is already working before introducing a technological solution.
- Respect Indigenous, traditional and local knowledge.
- Work toward community-led and community-controlled outcomes.

Paraphrased from Design Justice Network Principles in Sasha Costanza-Chock, “Design Justice, A.I., and Escape from the Matrix of Domination”, https://jods.mitpress.mit.edu/pub/costanza-chock/release/4

AI for Social Good (AI4SG)

Example AI4SG Applications



https://slideslive.com/38924046/invited-talk-integrating-learning-with-game-theory-for-societal-challenges
Watch: 0:00–3:08, 4:08–5:10, 6:01–8:16, 29:32–30:51

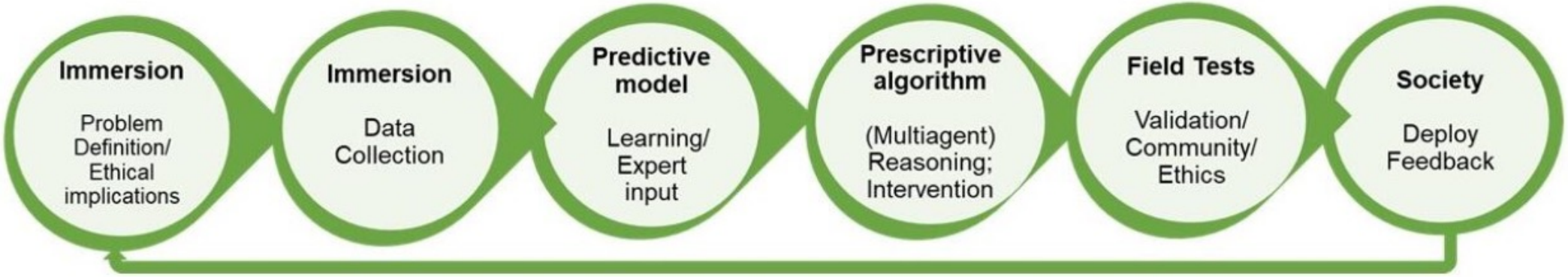
“AI for social impact” problem solving process (Milind Tambe)



As a first step, immersion in the domains is crucial to get a critical understanding of the problems, constraints and datasets. This may be a step that involves discussions with various stakeholders including the impacted community. In this step it is also important to build interdisciplinary partnerships and understand the challenges from the perspective of domain experts. Another aspect of this step is to understand data limitations and how to address these limitations.

Following an in-depth understanding via immersion, our next step is building a predictive model using machine learning or domain expert input; such a predictive model may for example predict which are high risk vs low risk cases in a population. Next, the prescriptive algorithm phase that plans interventions -- I will use the example of “game theoretic reasoning” but it could be any relevant intervention. Work on AI for social impact is often focused on domains where access to data is difficult (e.g., low resource communities or emerging market countries), and thus the challenge is often to plan interventions despite data that is uncertain and sparse.

Finally, we are also keenly interested in the final step of field testing and deployment. This is not only because we are interested in the social impact, but because these tests are often how we can learn key limitations of our models and algorithms, often leading to fundamental new research challenges to address. This research critically requires interdisciplinary partnerships for immersion and field testing; often we find such interdisciplinary partnerships lead to new research that is outside the scope of any one discipline.



Source: https://projects.iq.harvard.edu/cs288 (2024)

If you want to dig deeper

Even if computing is not ethically neutral, surely mathematics is!

Not according to this podcast:

- “What on Earth is Ethical Maths? With Maurice Chiodo”, The Good Robot Podcast, https://www.thegoodrobot.co.uk/listennow/episode/7e08a028/what-on-earth-is-ethical-maths-with-maurice-chiodo
- Also, see https://en.wikipedia.org/wiki/Ethics_in_mathematics